

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning



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Related SDGs:

SDG No. 9: Industry, Innovation, and Infrastructure

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Mini Project Presentation 2-B

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PROBLEM STATEMENT

Subsurface Ice Enables Sustainable Lunar Habitation The Limitation of Past Discoveries:

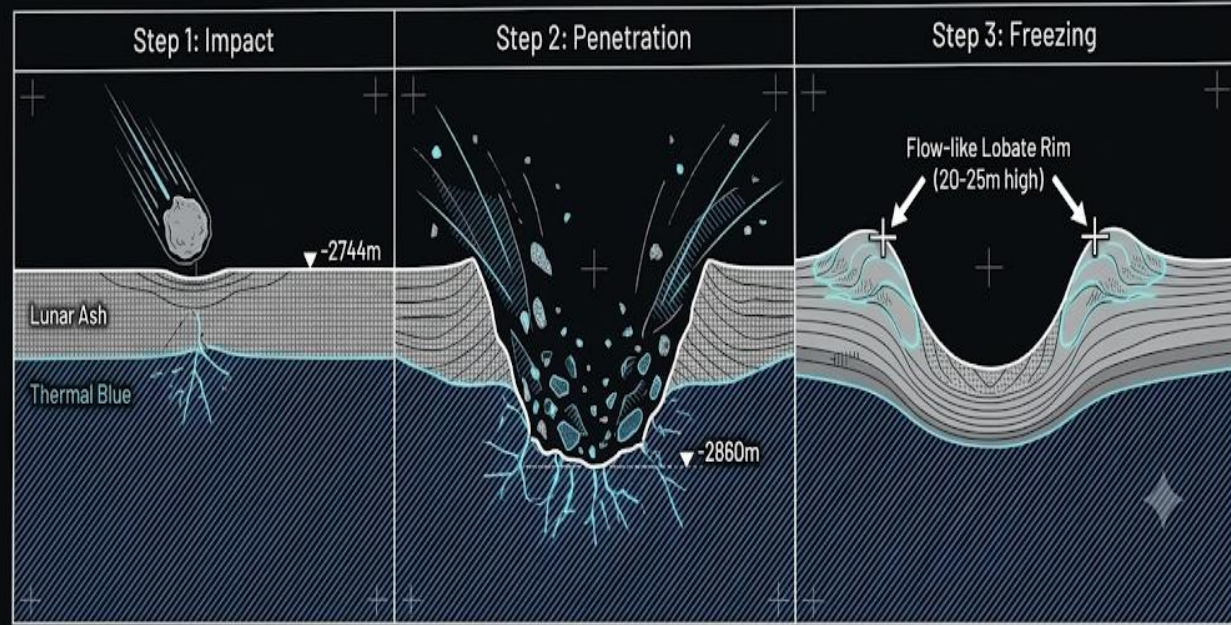
Surface Frost Earlier missions detected surface water molecules. However, surface frost is highly vulnerable to sublimation and solar radiation, making it difficult to harvest at scale.

The Subsurface Advantage:

Ice buried within the top few meters of regolith is protected from thermal swings. It is stable, mineable, and exists in quantities capable of sustaining permanent infrastructure.

The Lobate Rim Provides Geological Proof of Excavation

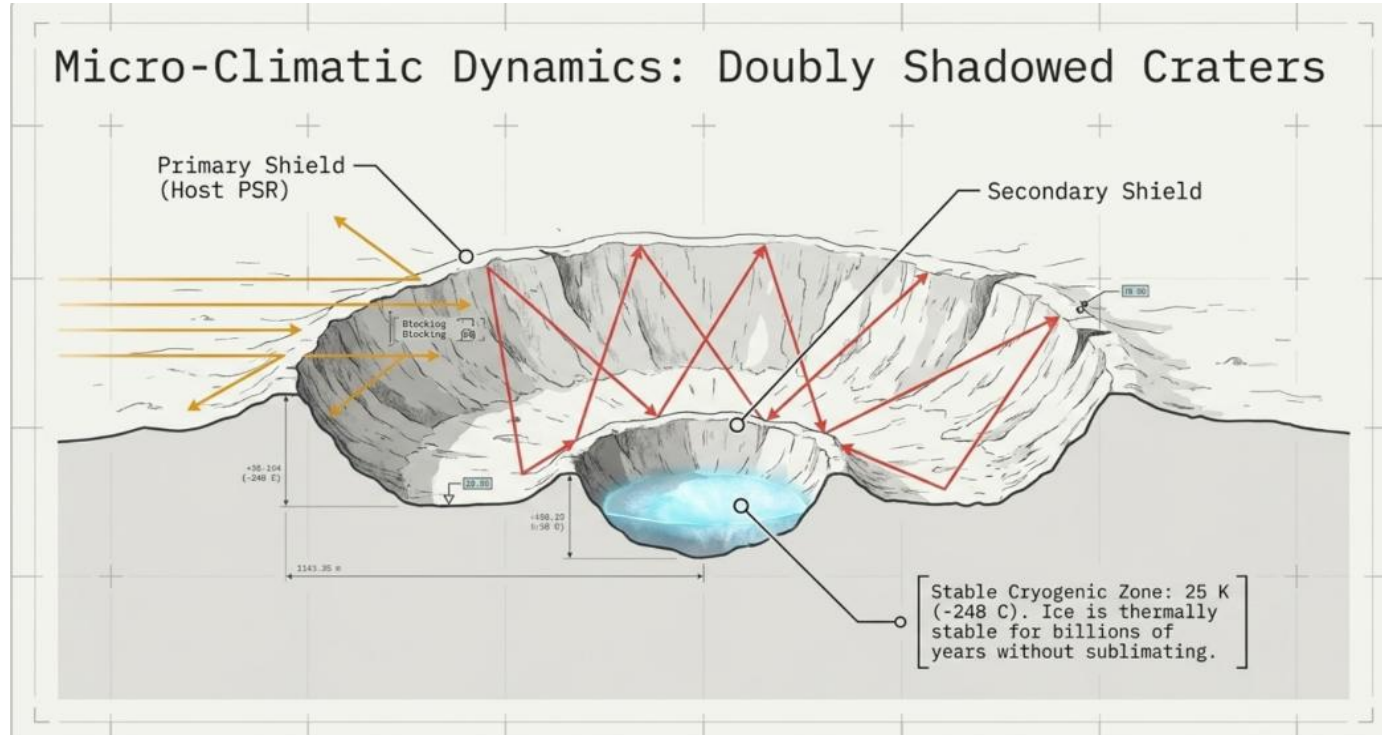
Radar data is invisible. The physical shape of F2 provides independent geological corroboration.



PROBLEM STATEMENT

Chandrayaan-2 DFSAR Pierces the Lunar Regolith:

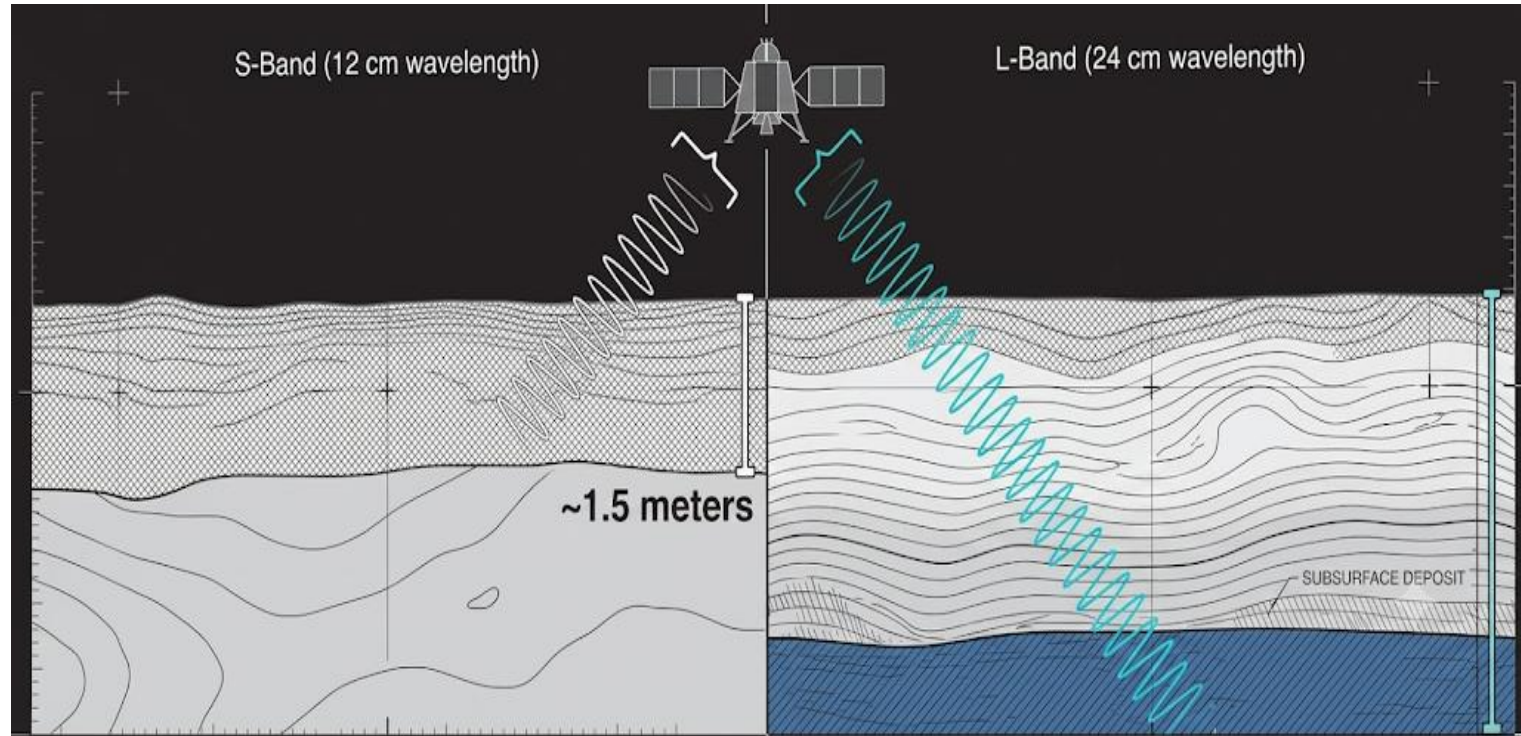
The Dual Frequency Synthetic Aperture Radar (DFSAR) is the first fully-polarimetric SAR to study the Moon, operating in both S-band and L-band frequencies.



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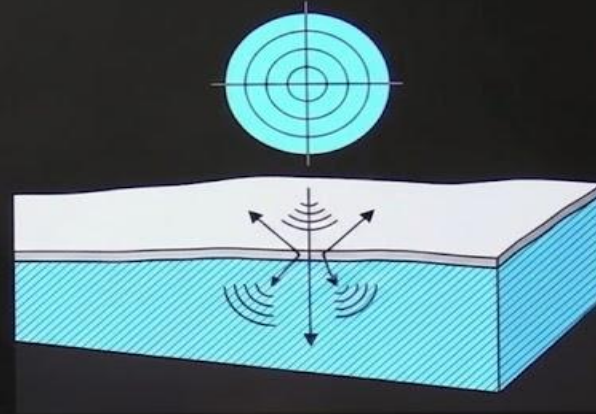
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The False Positive Problem of Rough Terrain

Elevated CPR (>1) is not exclusive to ice. Previous missions could not definitively separate fields of dry rocks from buried glaciers based on CPR alone.



Rough Rocks (Double-Bounce Reflection)



Subsurface Ice (Volumetric Scattering)

The Challenge: Rough, blocky surfaces like crater walls also produce $\text{CPR} > 1$.
A secondary metric is required to isolate genuine ice.

PROBLEM STATEMENT

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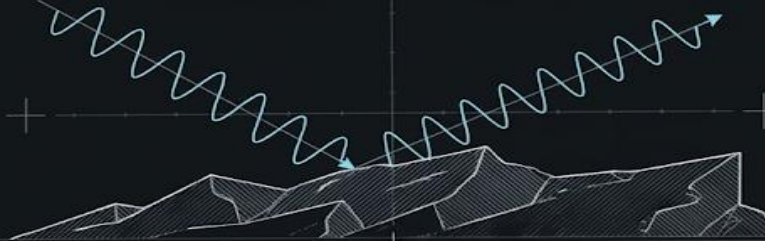
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Degree of Polarization (DOP) Separates Rock from Ice

DOP measures how much of the reflected radar signal retains its original, structured polarization state.

Path A: Rough Rocks

	1.00	2.00	3.20
	0.03	0.22	0.55
	0.02	1.00	0.25



High DOP (> 0.35).
Signal retains
structural integrity.

Path B: Subsurface Ice

	1.00	2.00	3.25
	0.09	0.02	0.55
	0.39	11.55	0.28



Low DOP (< 0.13).
Severe depolarization
due to random internal
scattering.

PROPOSED SOLUTION

How LAEP Solves the Problem

- **Module 1 (Detection):** Translates raw DFSAR physics into a continuous **Ice Confidence Score (0–1)** rather than a brittle binary classification.
- **Module 2 (Terrain Intelligence):** Integrates slope, roughness, and boulder data into a comprehensive hazard map to link detection with safety.
- **Module 3 (Landing Site Selection):** Ranks the top 5 safest, ice-proximate landing zones using a multi-objective **Landing Suitability Index**, treating safety as a hard gate.
- **Module 4 (Rover Traverse):** Deploys reachability-aware *A search algorithms** (NASA-standard) to map safe, energy-efficient routes that factor in hazards and shadows.
- **Module 5 (Resource Estimation):** Delivers an honest volume and mass *range* (accounting for real uncertainty) to support grounded cost/benefit decisions.

Unique Selling Proposition (USP)

- LAEP is the only pipeline that bridges raw radar physics directly to a fully justified, safety-gated, and reachability-checked mission plan with every decision traceable to published Chandrayaan-2 science (Putrevu et al., 2021).

The LAEP Differentiator (How We Stand Out)

- ❑ **End-to-End Mission Pipeline:** We go beyond basic single-band CPR thresholding. LAEP is a complete, actionable framework spanning detection, landing safety, rover routing, and volume estimation.
- ❑ **Dual-Frequency Consistency:** Leverages both L-band and S-band physical consistency to reliably distinguish true subsurface ice from surface rocks.
- ❑ **Unsupervised ML Validation:** Uses an Isolation Forest anomaly layer to cross-validate physics-based detections, overcoming the complete lack of ground-truth labels that limits standard supervised ML.



IF (Pixel_CPR > 1.0) AND (Pixel_DOP < 0.13) = SUBSURFACE ICE

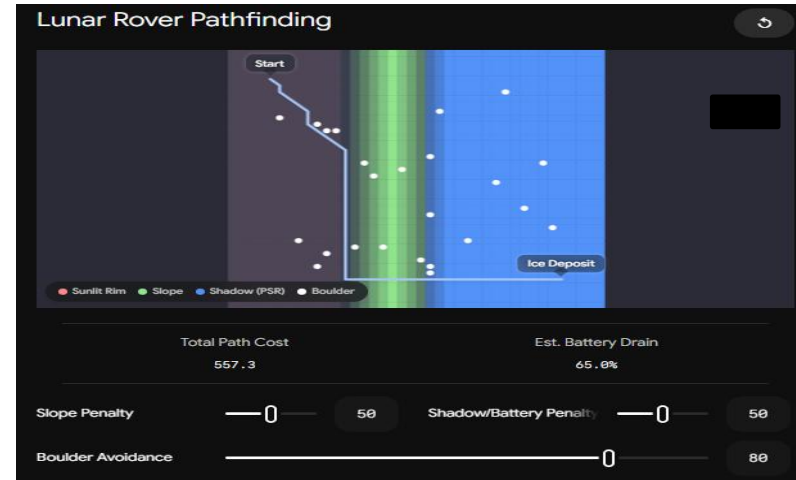
Scope & Objectives

List of features offered by the solution:

- ❖ **Physics-grounded Ice Confidence Score (ICS)** — combines CPR, DOP, backscatter, and dual-band (L/S) consistency into one continuous 0–1 score per pixel, fully interpretable and defensible (no black-box labels required).
- ❖ **Three-class terrain hazard mapping** — slope, roughness, and boulder density fused into Safe / Moderate / Dangerous zones, used as a hard safety gate downstream.
- ❖ **Multi-objective Landing Site Selection** — ranks the top-5 candidate landing zones (LZ-1 to LZ-5) by a weighted Landing Suitability Index, with non-maximum suppression so candidates aren't redundant near-duplicates.
- ❖ **Reachability-aware A* rover path planning** — computes the lowest-risk (not just shortest) traverse from landing site to ice deposit, pre-filtered by BFS reachability so it never targets physically inaccessible ice.
- ❖ **Ice volume & mass estimation with honest uncertainty bounds** — reports conservative / best-estimate / optimistic volume and mass, rather than a single over-confident figure.
- ❖ **Modular, real-data-ready architecture** — every "load" function is isolated in one section, so swapping simulated arrays for real ISSDC Pradan
- ❖ **Label-free ML cross-validation layer** — an Isolation Forest anomaly detector independently confirms physics-based ice detections without needing any ground-truth training labels.

DFSAR/OHRC/TMC-2 rasters requires no changes to the detection, planning, or estimation logic.

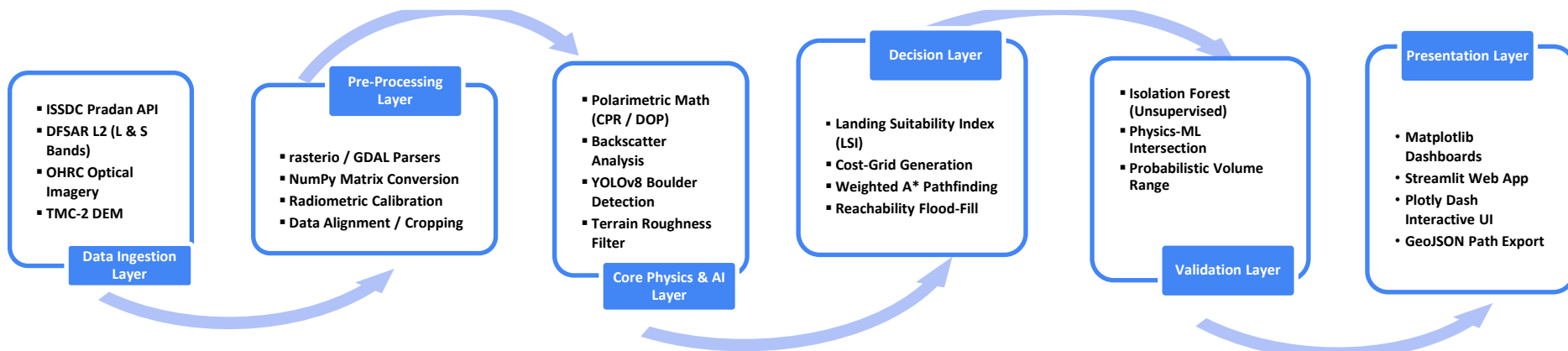
❖ **Demo-ready mission dashboard** — a single 6-panel visualization (DEM, ICS map, hazard map, landing candidates, rover path, and volume summary) built for direct portability into a Streamlit mission-planning interface.



How the Algorithm Mindset Shifts

- **On Earth:** The algorithm asks: *"Which combination of roads gets me there fastest while avoiding heavy traffic?"*
- **On the Moon:** The algorithm asks: *"Which exact sequence of physical steps lets me crawl down this hill without rolling over, avoids that boulder cluster, and keeps me in the sunlight long enough to avoid freezing?"*

Architecture diagram of the proposed solution



1. Data Ingestion Layer:

- Action:** Fetches raw, multi-sensor satellite data (DFSAR radar, OHRC optical, TMC-2 DEM) from ISRO repositories into a central staging environment.
- Outcome/Pros:** Centralizes all raw telemetry, ensuring the downstream pipeline has instant access to a unified, uncorrupted, and synchronized single source of truth.

2. Pre-Processing Layer:

- Action:** Uses GDAL/rasterio to align and convert raw .TIF/.IMG files into standardized 25-meter NumPy matrices for fast calculation.
- Outcome/Pros:** Eliminates spatial coordinate mismatches entirely and dramatically accelerates all subsequent heavy mathematical matrix computations.

3. Physics & AI Engine:

- Action:** Calculates radar physics (CPR/DOP) to locate subsurface ice and applies YOLOv8 object detection to map surface boulder hazards.
- Outcome/Pros:** Creates a highly accurate, dual-layered understanding of the environment, instantly separating scientific targets (ice) from physical threats (boulders).

4. Mission Decision Layer:

- Action:** Merges hazard data into a Cost-Grid to rank safe landing sites (LSI) and plots the safest rover route using A* search algorithms.
- Outcome/Pros:** Translates raw science into strict engineering constraints, guaranteeing that the proposed lander coordinates and rover paths are physically survivable.

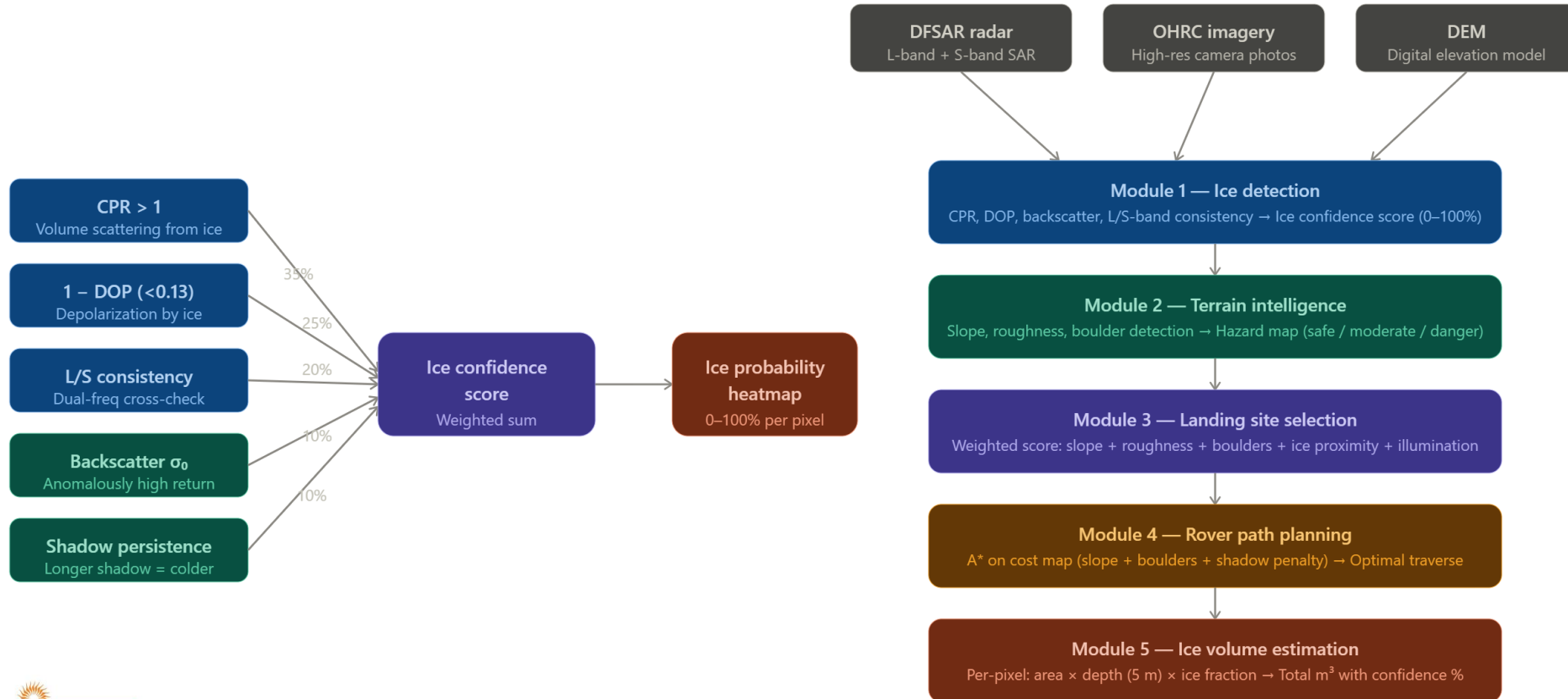
5. Validation Layer:

- Action:** Independently cross-verifies the physics-based ice detections using an unsupervised ML model (Isolation Forest) to ensure scientific accuracy.
- Outcome/Pros:** Delivers a scientifically defensible, label-free guarantee of ice presence, fully compensating for the lack of ground-truth training data on the lunar surface.

6. Presentation Layer:

- Action:** Translates the final data into interactive web dashboards (Streamlit/Plotly) and exportable mission maps (GeoJSON/PDF) for human commanders.
- Outcome/Pros:** Transforms millions of complex data points into a clear, intuitive, and immediately actionable mission briefing that stakeholders can effortlessly approve.

Process flow diagram or Workflow



TECH STACK

Category	Technology / Library	Specific Application in LAEP	Technical Justification
Core Computation	NumPy	All Modules (Data foundation)	Handles all raster data as 2D arrays (np.ndarray). Ensures vectorized, high-speed mathematical operations for physics formulas.
Spatial Processing	SciPy (scipy.ndimage)	Module 2 & 3: Terrain & Site Selection	Executes spatial filtering like generic_filter for terrain roughness, and distance_transform_edt for precise distances to ice targets.
Machine Learning	Scikit-Learn (IsolationForest)	Validation Layer: Anomaly Detection	Provides an unsupervised ensemble algorithm to detect statistical anomalies in the radar/terrain stack without requiring ground-truth labels.
Pathfinding & Structures	Python Std Lib (heapq, deque)	Module 4: Rover Path Planning	heapq implements the min-heap priority queue for the A* search; deque executes efficient Breadth-First Search (BFS) for reachability.
Visualization	Matplotlib	Outputs & Mission Dashboard	Generates the 6-panel final dashboard, applying specific colormaps and rendering the 2D geospatial heatmaps and path overlays.
Data Ingestion	Rasterio / GDAL	Section 0: Real Data Integration	Required for the next phase to parse actual geospatial formats (GeoTIFF, IMG) directly from the ISSDC Pradan portal into NumPy arrays.
Deployment	Streamlit / Plotly Dash	Live Hackathon Demo / UI	Will rapidly convert the pure Python backend into an interactive, browser-based web application for judges to tweak parameters in real-time.

ALGORITHM

ALGORITHM: Lunar Autonomous Exploration Pipeline (LAEP)

INPUTS:

DFSAR_L, DFSAR_S // L-band and S-band Radar Arrays (CPR, DOP, Backscatter)
OHRC_DEM // Digital Elevation Model (3D topography)
OHRC_IMG // High-resolution optical imagery

OUTPUTS:

LZ_Optimum // (x, y) coordinates of the best landing site
Rover_Path // Array of (x, y) coordinates from LZ to Ice
Ice_Metrics // Validated volume and mass of ice deposit

// --- PHASE 1: PHYSICS-BASED ICE DETECTION ---

FUNCTION Detect_Ice(DFSAR_L, DFSAR_S):

Initialize Ice_Map as 2D Array

FOR EACH pixel (i, j) IN DFSAR_L:

cpr = Calculate_CPR(pixel)

dop = Calculate_DOP(pixel)

ls_diff = AbsoluteDifference(DFSAR_L[i,j], DFSAR_S[i,j])

// Strict physics filter

IF cpr > 1.0 AND dop < 0.13:

Ice_Map[i,j].score = (0.35 * cpr) + (0.25 * (1-dop)) + (0.20 * (1-ls_diff))

ELSE:

Ice_Map[i,j].score = 0

RETURN Ice_Map

// --- PHASE 2: TERRAIN INTELLIGENCE & HAZARD MAPPING ---

-

FUNCTION Build_Cost_Grid(OHRC_DEM, OHRC_IMG):

Initialize Cost_Grid as 2D Array

FOR EACH pixel (i, j) IN OHRC_DEM:

slope = Compute_Gradient(OHRC_DEM[i,j])

roughness = Compute_Local_Variance(OHRC_DEM[i,j], window=5)

boulders = YOLOv8_Detect_Boulders(OHRC_IMG[i,j])

// Hard safety constraints

IF slope > 15° OR boulders > THRESHOLD:

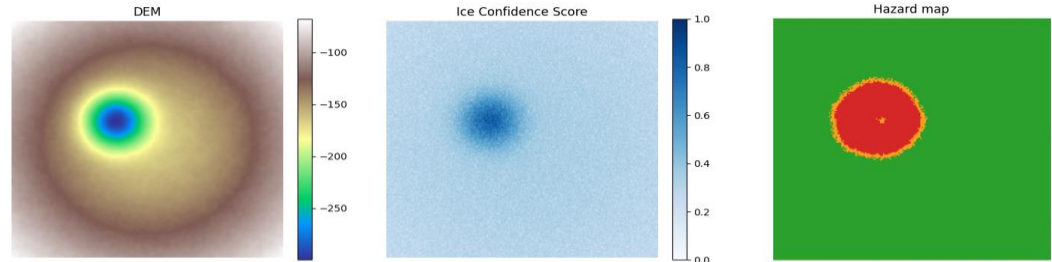
Cost_Grid[i,j] = INFINITY // Impassable No-Go Zone

ELSE:

Cost_Grid[i,j] = Base_Cost + (W1 * slope) + (W2 * roughness)

RETURN Cost_Grid

Lunar Autonomous Exploration Pipeline — Mission Dashboard



ALGORITHM

LAEP ALGORITHM

// --- PHASE 3: LANDING SITE SELECTION ---

```
FUNCTION Optimize_Landing(Cost_Grid, Ice_Map, Illumination_Map):  
    Initialize LSI_Scores as List  
    Safe_Zones = Extract_Pixels(Cost_Grid < INFINITY AND Illumination_Map  
> 80%)  
    FOR EACH zone IN Safe_Zones:  
        distance_to_ice = Euclidean_Distance(zone, Max(Ice_Map.score))  
        LSI = (0.4 * (1 - Cost_Grid[zone])) + (0.6 * (1 - distance_to_ice))  
    APPEND zone TO LSI_Scores  
  
LZ_Optimum = Pop_Maximum(LSI_Scores) // Extract highest scoring site  
RETURN LZ_Optimum
```

// --- PHASE 4: ROVER PATH PLANNING (A* SEARCH) ---

```
FUNCTION Navigate_Rover(LZ_Optimum, Ice_Map, Cost_Grid):  
    Target_Ice = Coordinate of Max(Ice_Map.score)  
  
    // A* pathfinding using f(n) = g(n) + h(n)  
    Rover_Path = A_Star_Search(  
        Start = LZ_Optimum,  
        Goal = Target_Ice,  
        Grid = Cost_Grid,  
        Heuristic = Euclidean_Distance  
    )  
    RETURN Rover_Path
```

// --- PHASE 5: ML CROSS-VALIDATION & VOLUME ESTIMATION ---

```
FUNCTION Verify_And_Quantify(Ice_Map, DFSAR_L, OHRC_DEM):  
    // Unsupervised anomaly detection  
    ML_Anomaly_Mask = Isolation_Forest(Features: [CPR, DOP, Slope, Roughness])  
  
    // Intersect Physics with ML  
    Validated_Ice = Intersect(Ice_Map > 0.6, ML_Anomaly_Mask)  
  
    Ice_Metrics.Area = Count(Validated_Ice) * Pixel_Area  
    Ice_Metrics.Volume = Integrate(Validated_Ice * Depth=5m * Ice_Fraction=0.35)  
  
    RETURN Ice_Metrics
```

// --- MAIN EXECUTION ---

```
Ice_Map = Detect_Ice(DFSAR_L, DFSAR_S)  
Cost_Grid = Build_Cost_Grid(OHRC_DEM, OHRC_IMG)  
LZ_Optimum = Optimize_Landing(Cost_Grid, Ice_Map, Illumination_Map)  
Rover_Path = Navigate_Rover(LZ_Optimum, Ice_Map, Cost_Grid)  
Final_Report = Verify_And_Quantify(Ice_Map, DFSAR_L, OHRC_DEM)
```

PRINT Final_Report

MISSION SUMMARY

```
-----  
Best landing site: LZ-1  
Position: row 37, col 99  
LSI score: 0.924  
  
Rover traverse:  
Distance: 2.96 km  
Waypoints: 103  
  
Ice deposit:  
Area: 0.530 km^2  
Volume (best est): 649,197 m^3  
Mass equivalent: 595,314 tonnes  
  
Cross-validated confidence:  
848 px confirmed by  
both physics + ML
```

